

Course Number	COE 328
Course Title	Digital Systems
Semester/Year	Semester 3, Year 2
Section Number	1

Lab 4 - Circuit & Storage Elements

Assignment Title	Lab 4
Submission Date	th, 2022
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(Note: Remove the first 4 digits from your student ID)

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In-Lab:

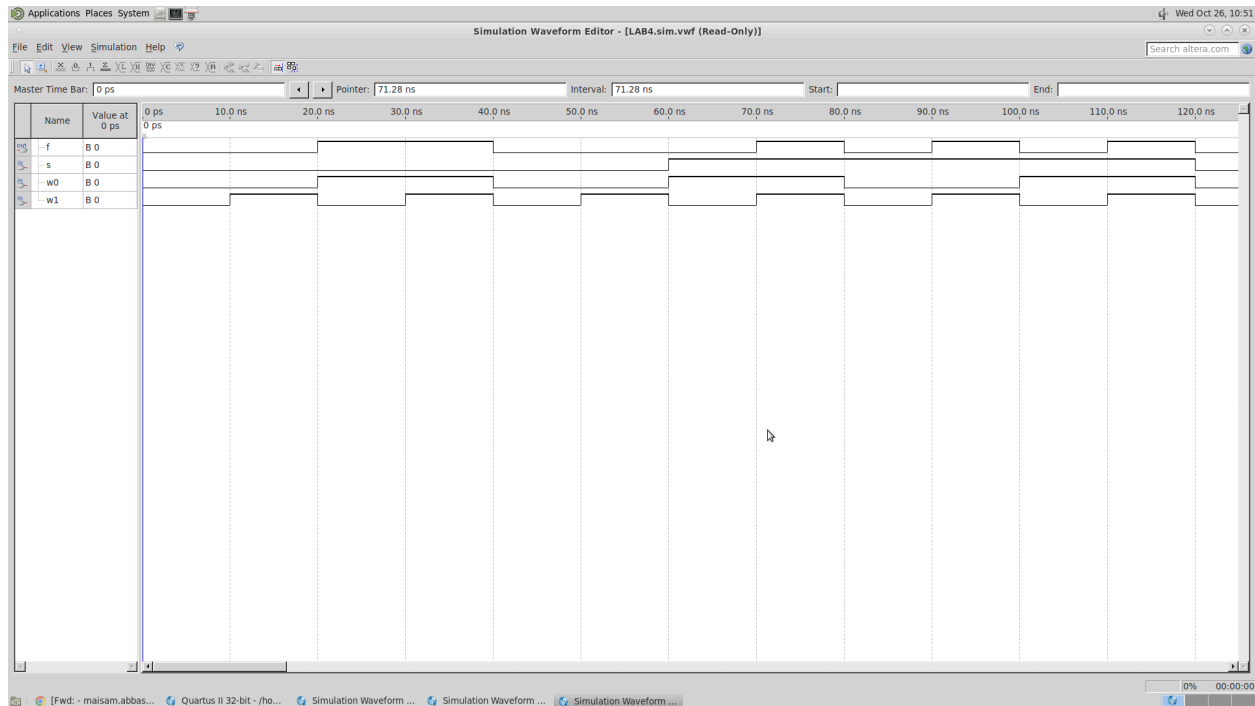


Figure 1: Waveform of 2-to-1 multiplexer

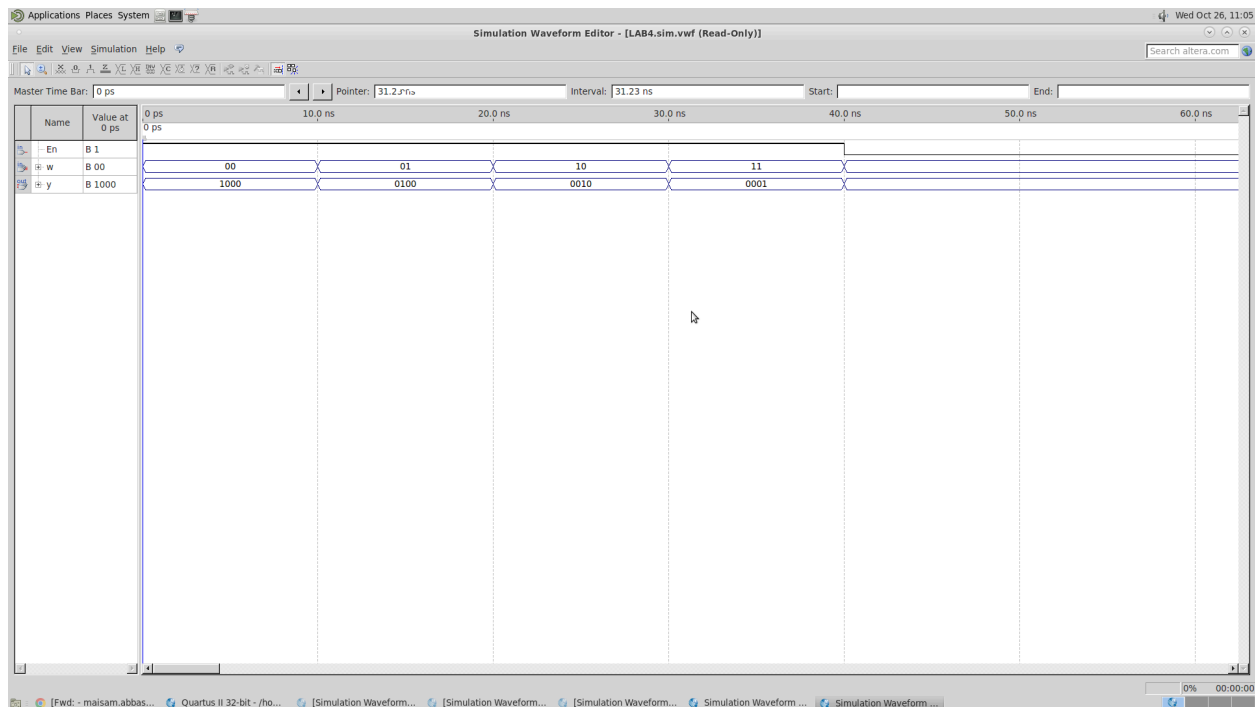


Figure 2: Waveform of 2:4 decoder

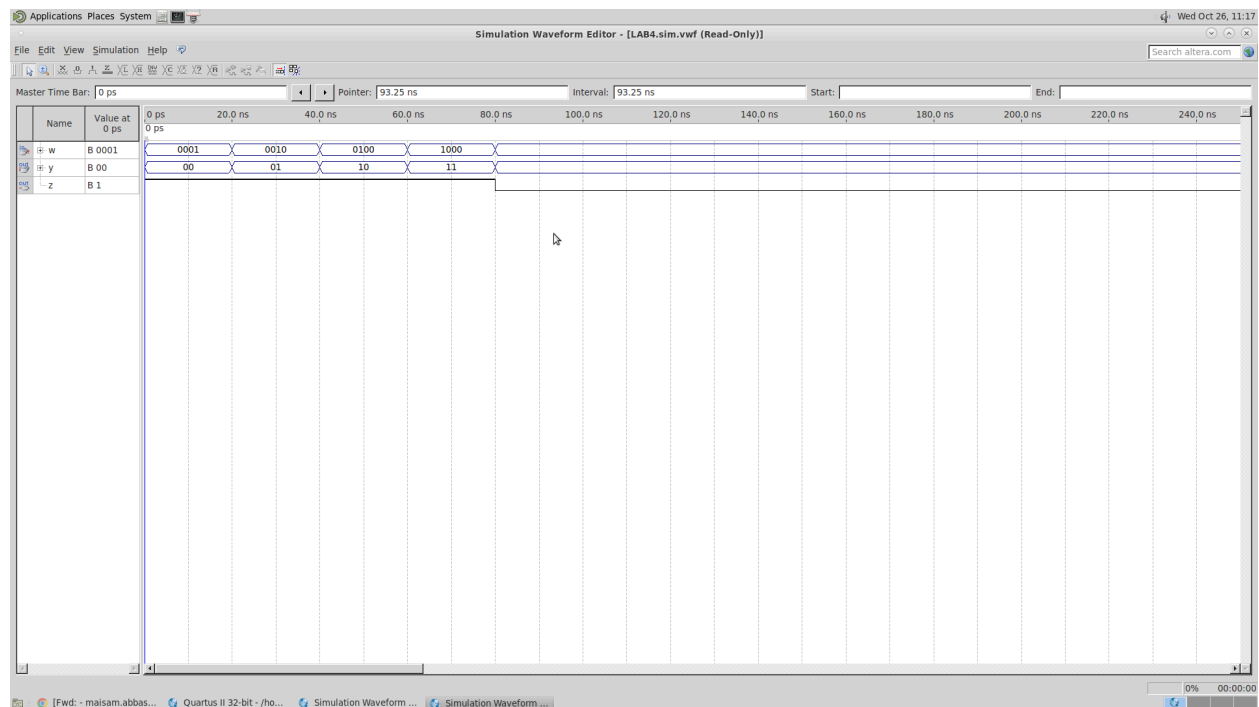


Figure 3: Waveform of priority encoder

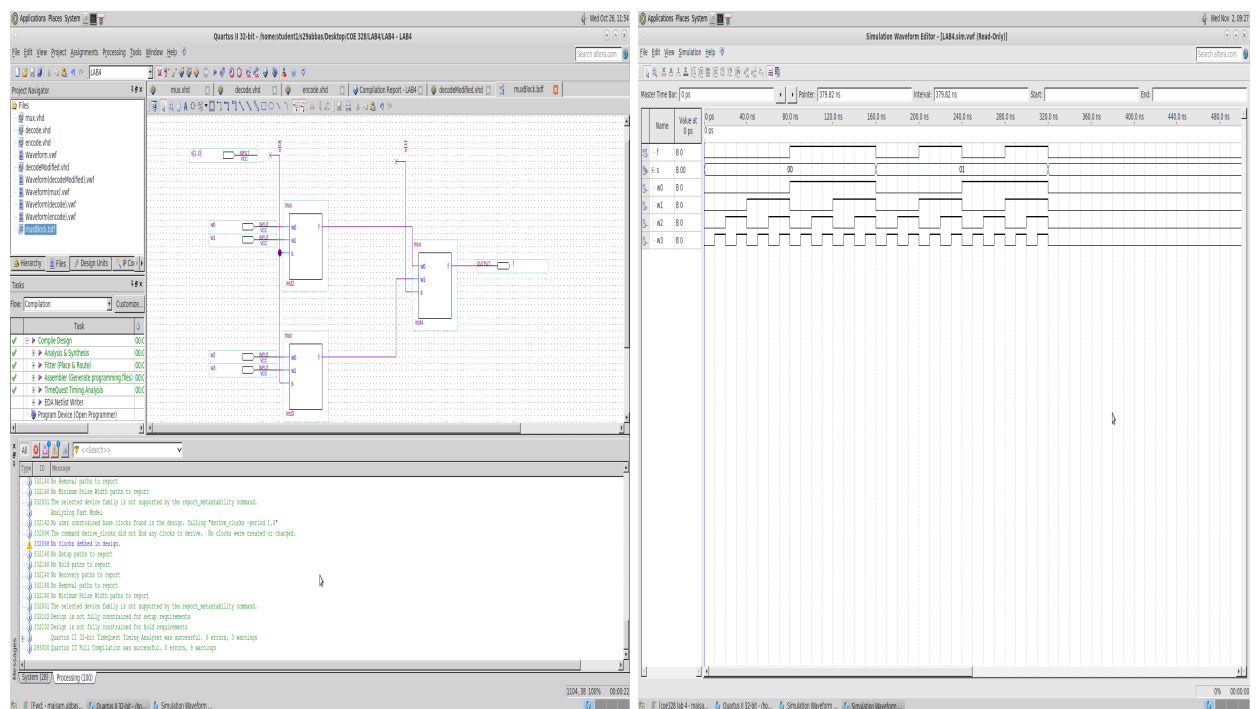


Figure 4 & 5: Block diagram and waveform of 4-to-1 multiplexer

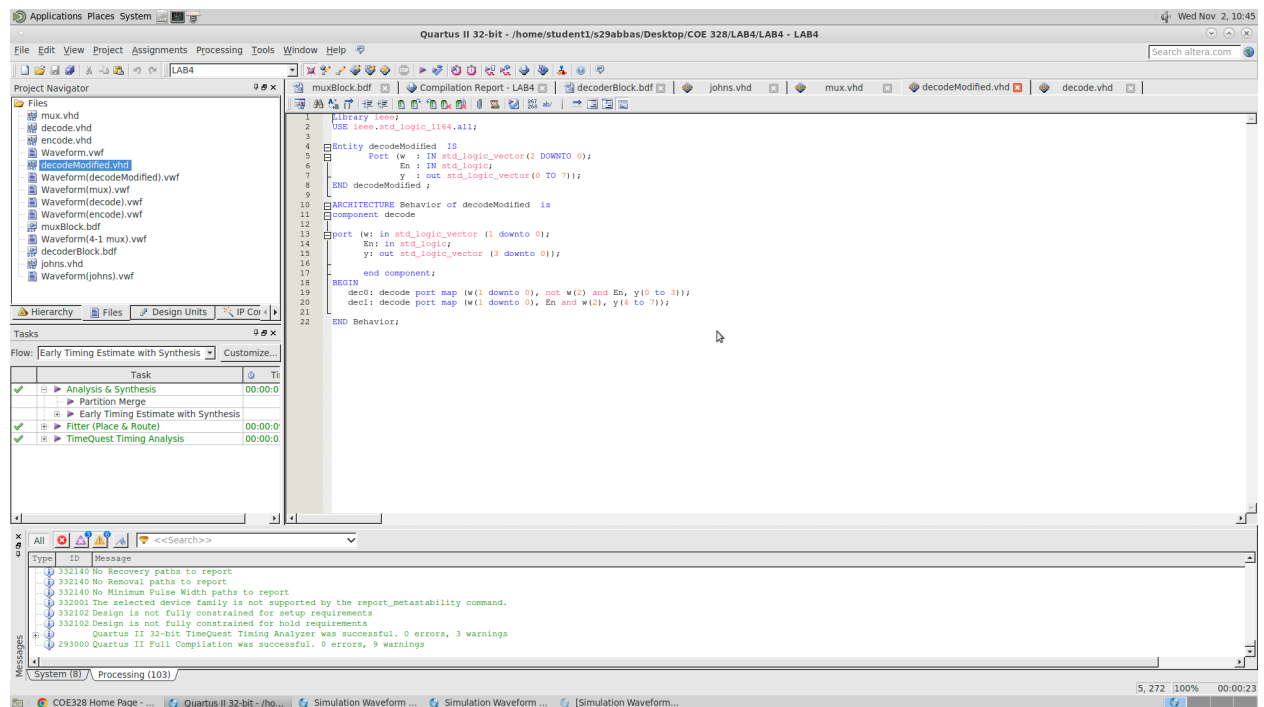


Figure 6: Modified VHDL code for decoder (2:4 to 3:8)

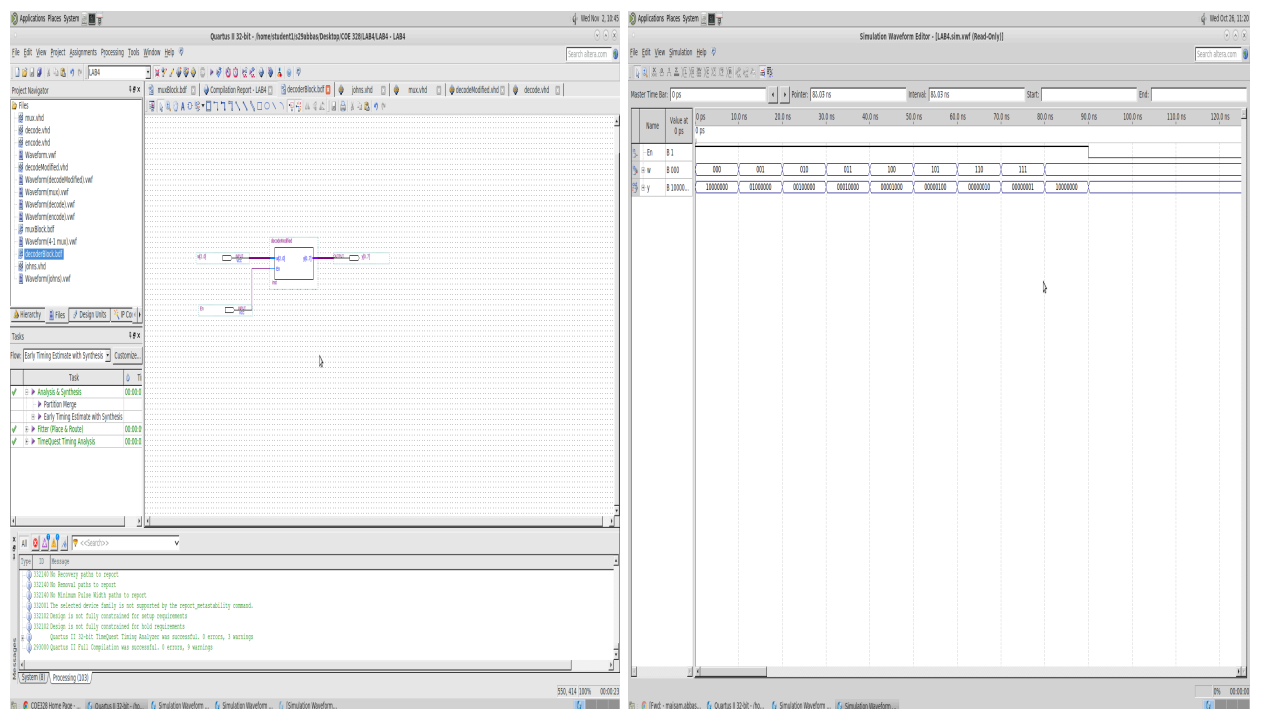


Figure 7 & 8: Block diagram and waveform of 3:8 decoder

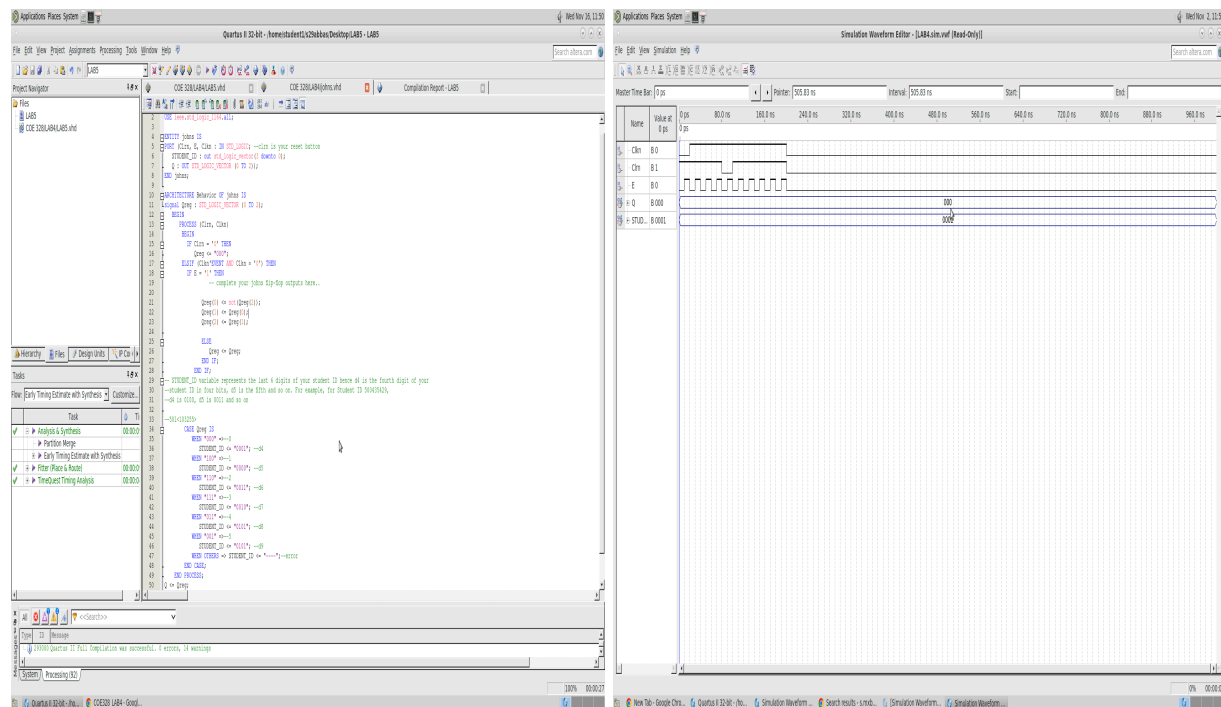


Figure 9 & 10: VHDL code and waveform for Johnson Counter

Using the VHDL code, the case code ensures that the 3-bit binary is converted to a 4-bit binary number to configure the required student number. Additionally, the output of the proceeding flip-flop is connected as the input of the next flip-flop. Therefore, this waveform represents the given student number.

	Q0	Q1	Q2
0	0	0	0
1	1	0	0
2	1	1	0
3	1	1	1
4	0	1	1
5	0	0	1

Table 1: John's truth table

Student ID: 501103255

0101 0000 0001 0001 0000 0011 0010 0101 0101
 1 0 3 2 5 5

last 6 digits of Student number

$x_3 x_4$	00	01	11	10
$x_1 x_2$	00	1	1	1
	01			
	11			
	10			

$$f(x_1, x_2, x_3, x_4) = m\sum(1, 0, 3, 2, 5)$$

$$f = \bar{x}_1 \bar{x}_2 + \bar{x}_1 \bar{x}_3 x_4$$

$$g = \bar{x}_1 \bar{x}_2, \quad h = \bar{x}_1 \bar{x}_3 x_4$$

• Truth table

m	x_1	x_2	x_3	x_4	g	h	f
0	0	0	0	0	1	0	1
1	0	0	0	1	1	1	1
2	0	0	1	0	1	0	1
3	0	0	1	1	1	1	1
4	0	1	0	0	0	0	0
5	0	1	0	1	0	0	0
6	0	1	1	0	0	0	0
7	0	1	1	1	0	0	0
8	1	0	0	0	0	0	0
9	1	0	0	1	0	0	0
10	1	0	1	0	0	0	0
11	1	0	1	1	0	0	0
12	1	1	0	0	0	0	0
13	1	1	0	1	0	0	0
14	1	1	1	0	0	0	0
15	1	1	1	1	0	0	0

$$g = \bar{x}_1 \bar{x}_2$$

$$h = \bar{x}_1 \bar{x}_3 x_4$$

$$f = \bar{x}_1 \bar{x}_2 + \bar{x}_1 \bar{x}_3 x_4$$

Figure 11: K-map and truth table for student ID